# Antimicrobial agents inhibiting bacterial Nucleic Acid Synthesis

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# Nucleic Acid Synthesis inhibitors

## FLUOROQUINOLONES

- Ciprofloxacin
- Levofloxacin
- Moxifloxacin
- Nalidixic acid
- Norfloxacin
- Ofloxacin

## FOLATE ANTAGONISTS

### INHIBITORS OF FOLATE SYNTHESIS

- Mafenide
- Silver sulfadiazine
- Sulfasalazine

### INHIBITORS OF FOLATE REDUCTION

- Pyrimethamin
- Trimethoprim

# Fluoroquinolones

Fluoroquinolones, are a class of man-made antibiotics. They are synthetic analogue of Nalidixic acid.

Fluoroquinolones have broad action against aerobic gram-negative bacilli they are bactericidal and exhibit Conc. dependent killing activity ( **post-antibiotic effect)**

**Members:**

**1st gen.:** Nalidixic acid

**2ndgen**.: Ciprofloxacin (high affinity for DNA gyrase topoisomerase II) , Norfloxacin

**3rd gen**.: Levofloxacin

**4th gen.:** Moxifloxacin (high affinity for topoisomerase IV)

Levofloxacin and moxifloxacin are sometimes referred to as "respiratory fluoroquinolones," because they have excellent activity against *Streptococcus pneumoniae*, which is a common cause of community-acquired pneumonia (CAP).

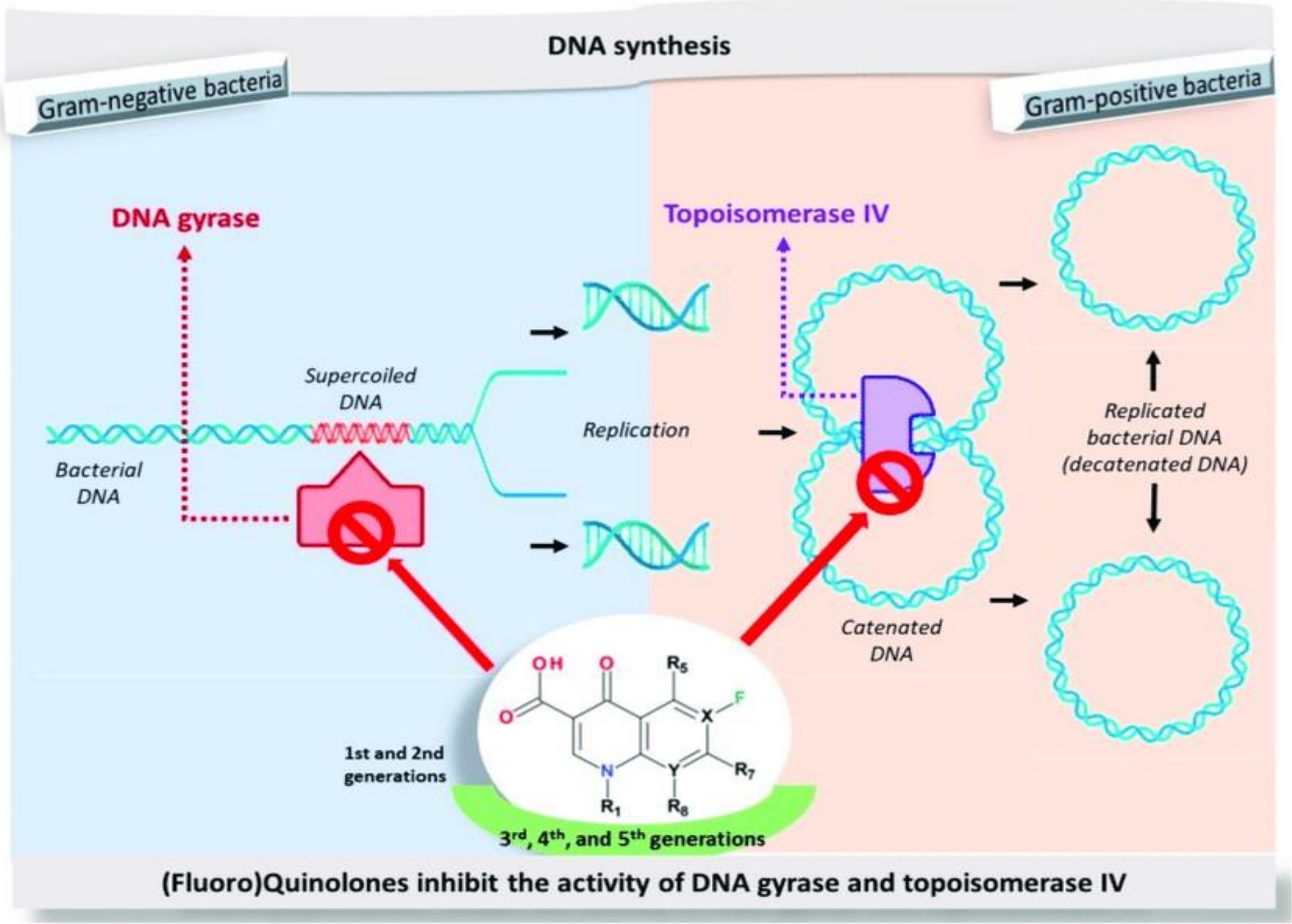
## Mechanism of Action:

Fluoroquinolones enter bacteria through porin channels and exhibit antimicrobial effects by:

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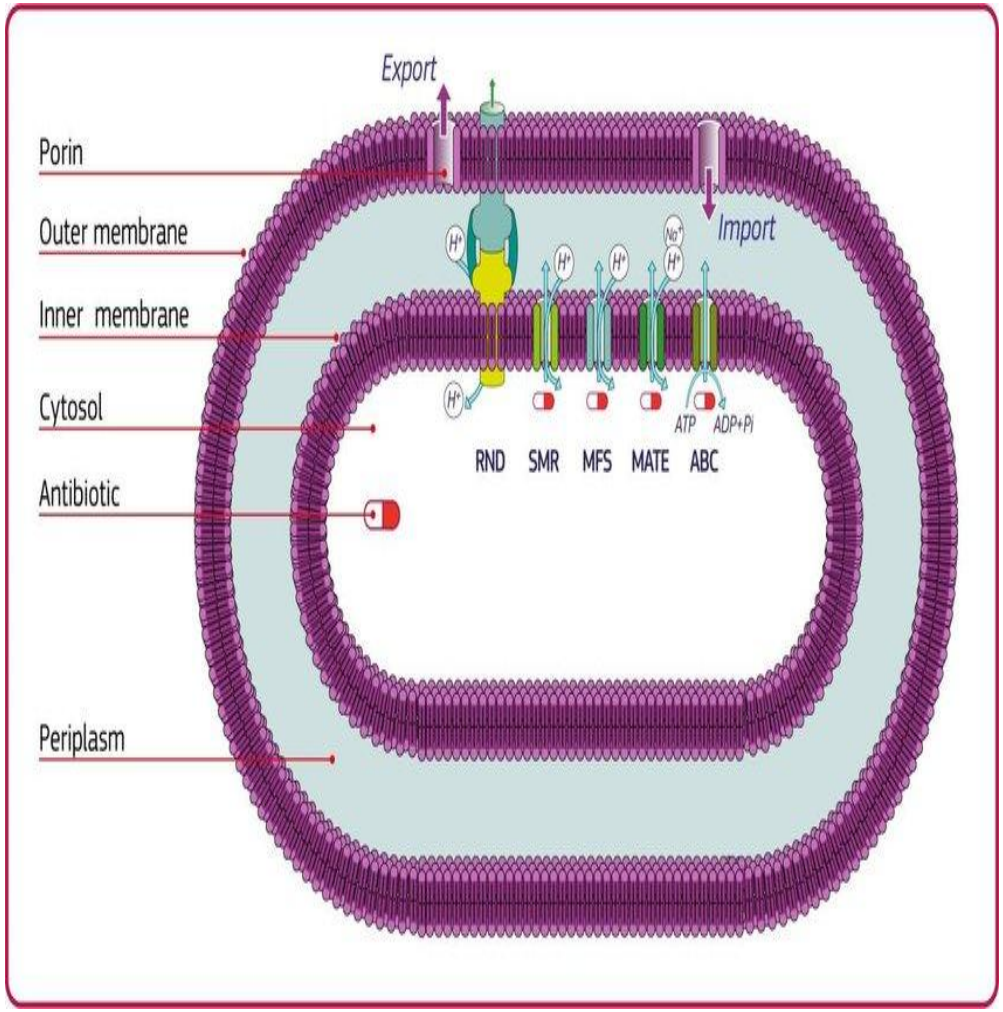
1. Inhibiting DNA gyrase (bacterial topoisomerase II) which results in relaxation of supercoiled DNA, promoting DNA strand breakage (more in gram-negative organisms, e.g. *Pseudomonas aeruginosa*)

2. Inhibiting bacterial topoisomerase IV: affects chromosomal stabilization during cell division, thus interfering with the separation of newly replicated DNA (more in gram-positive organisms, e.g. *Streptococcus pneumoniae*




(Fluoro)Quinolones inhibit the activity of DNA gyrase and topoisomerase IV

**Resistance**: primarily due to chromosomal mutations
Both topoisomerase IV and DNA gyrase may undergo mutations.
Reduced intracellular concentration due to
(1). decreased number of porin proteins
(2). efflux pumps



## Pharmacokinetics:

- The quinolones are rapidly and almost completely absorbed after oral administration. Antacids and dietary supplements containing iron and zinc  interfere with its absorption
- Widely distributed in all body tissues with poor CSF penetration. **Levofloxacin** and **Moxifloxacin**  accumulate in neutrophils and macrophages, thus having activity against intracellular organisms.
- All quinolones are metabolized by hepatic cytochrome P450 enzymes and they are **enzyme inhibitors**. (e.g., Cipro) can raise other drugs levels in plasma by inhibiting their metabolism – e.g., theophylline, warfarin, caffeine and cyclosporine..

They are mainly excreted unchanged by kidneys ,so they are useful in UTIs (reduce dose in renal impairment). **Moxifloxacin** is excreted primarily by the liver, and no dose adjustment is required for renal impairment.

## Adverse effects:

**Most common:**

GIT: nausea, vomiting, and diarrhea

CNS: Headache, dizziness, insomnia

**Less common, but potentially severe:**

- Phototoxicity – avoid excessive sunlight
- Prolonged Q-T interval (palpitations, atrial flutter, ectopic beats, fainting, chest pain)
- Hypoglycemia (contraindicated in diabetics)
- Predispose to seizures in patients with history of epilepsy
- **Cartilage damage** (contraindicated in pregnancy and children <18 yrs)

## Clinical indication:

**Norfloxacin:** is effective in treating non-systemic infections, such as UTIs, prostatitis, and infectious diarrhea.

**Ciprofloxacin:** effective in the treatment of many systemic infections caused by gram-negative bacilli .

- Active against P. aeruginosa (in Cystic fibrosis).
- Traveler's diarrhea caused by E. coli
- Typhoid fever caused by Salmonella typhi.
- Used as second-line agent in the treatment of tuberculosis
- For treatment of both complicated and uncomplicated UTI
- is the drug of choice for post exposure prophylaxis and for treatment of anthrax
- It is not the drug of first choice for pneumonia or sinusitis, because it has weak activity against S. pneumoniae, a common causative agent

**Levofloxacin:**

- It has 100% bioavailability and is dosed once daily.
- It has broad spectrum of activity used in a wide range of infections, including prostatitis, skin infections, community-acquired pneumonia (CAP), and nosocomial pneumonia.
- Used in infections unresponsive to b-lactam antibiotics, such as amoxicillin-clavulanic acid.
- Levofoxacin and moxifoxacin are known as “respiratory fluoroquinolones” due to their excellent activity against S. pneumoniae

**Moxifloxacin:**

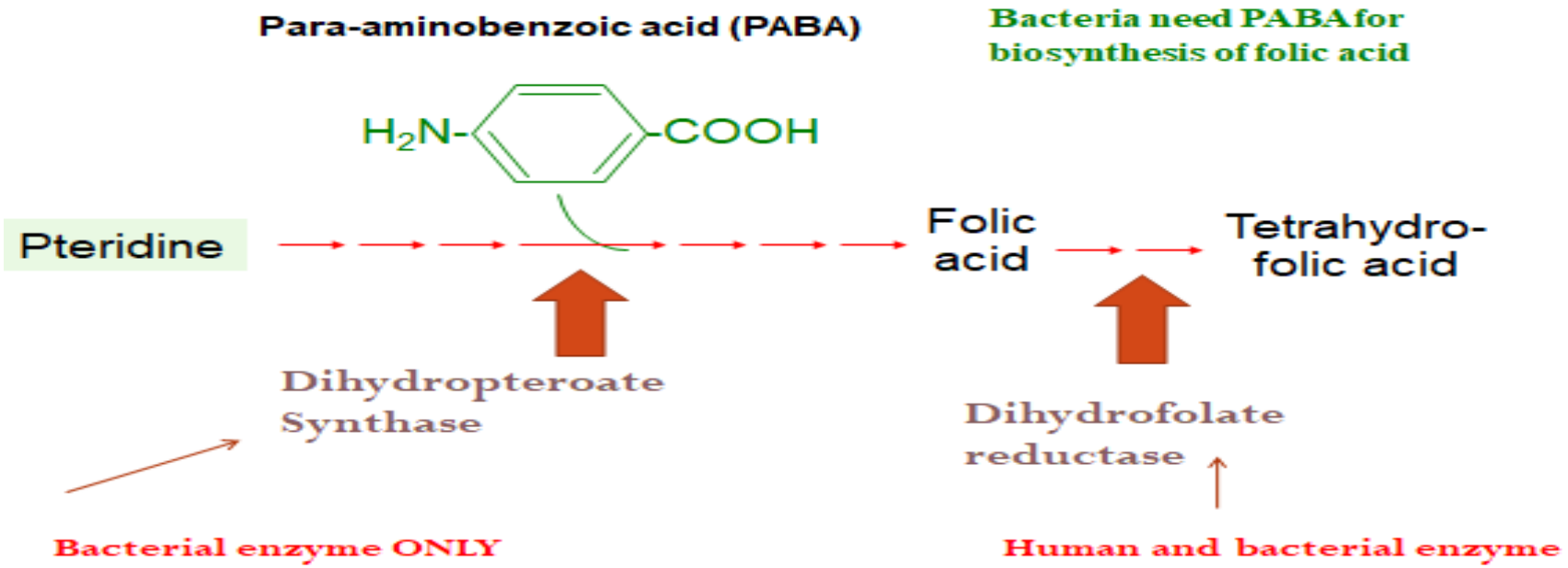
- Also has excellent activity against many anaerobes, ( resistant to Bacteroides fragilis).
- It has poor activity against P. aeruginosa.
- Does not concentrate in urine and not used for UTIs

# FOLATE ANTAGONISTS
# Sulfonamide and Trimethoprim

**Tetrahydrofolate (THF):** is a coenzyme required for the synthesis of purines and pyrimidines (precursors of RNA and DNA) in bacteria and humans.

To synthesize tetrahydrofolate humans must first obtain preformed folate in the form of folic acid from the diet.

In contrast, many bacteria are impermeable to folic acid and other folates and, therefore, they use P-aminobenzoic acid (PABA) to synthesize folate and then THF.



# Sulfonamides

Sulfonamides are a family of antibiotics that inhibit growth of bacteria by inhibiting synthesis of folate. They are bacteriostatic when given alone.

**Mechanism of action:** they are structural analogues of *p* - aminobenzoic acid because of their structural similarity to PABA, they compete with this substrate for the bacterial enzyme, **dihydropteroate synthetase** thus, they inhibit the synthesis of bacterial **dihydrofolic acid** and then **THF.**

**Resistance:** is generally irreversible and may be due to

**1)** an altered dihydropteroate synthetase,

**2)** decreased cellular permeability to sulfa drugs.

**3)** enhanced production of the natural substrate, PABA**.**

## Pharmacokinetics:

- Most sulfa drugs are well absorbed from GI tract except sulfasalazine( used for treatment of chronic inflammatory bowel disease)
- Binds variably to serum albumin.
- Distribute through out body fluids, and well penetrate CSF, also pass the placental barrier
- Metabolized in the liver by N-acetylation and then excreted in urine. The acetylated product have potential to precipitate at neutral or acidic pH of urine and causes crystalluria (stone formation).
- Sulfonamides may also be eliminated in breast milk.
- They are strong **inhibitor of CYP2C9** and potentiate the effect of oral anticoagulants, hypoglycemic agents, methotrexate, etc.

## Adverse effects:

**Crystalluria:** may lead to Nephrotoxicity (alkalinization of urine reduce crystaluria)

**Hypersensitivity:** such as rashes, angioedema or Stevens-Johnson syndrome, may occur

**Hematopoietic disturbances:**

Hemolytic anemia -in patients with (G6PD) deficiency

Agranulocytosis, aplastic anemia, and other blood dyscrasias may occur.

**Kernicterus:** This disorder may occur in newborns, because sulfa drugs displace bilirubin from binding sites on serum albumin

**Sulfonamides are contraindicated in newborns & during the last 2 months of pregnancy**

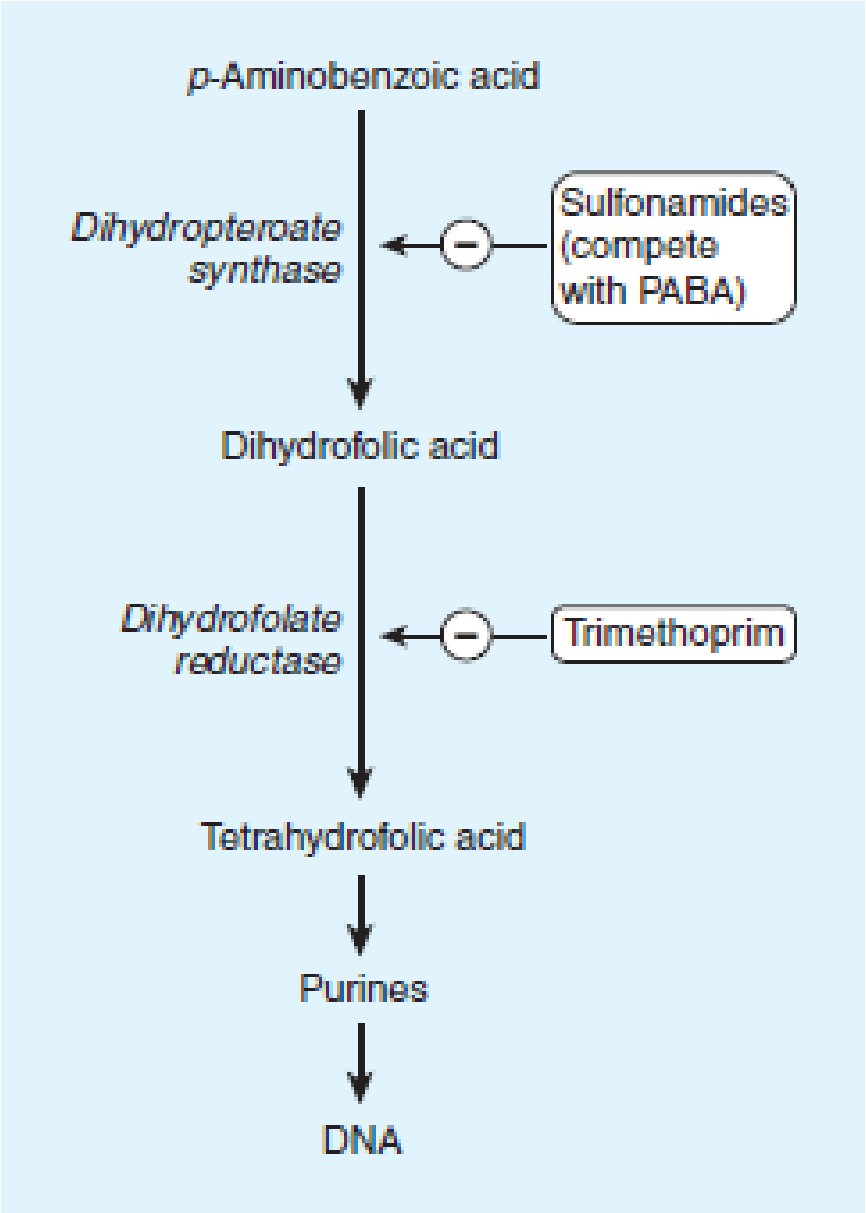
# Therapeutic Uses of Sulfonamides

- Sulfa drugs are active against selective Enterobacteriaceae in UTIs
- sulfadiazine  with pyrimethamine used in Toxoplasmosis (avoid using in pregnant women)
- Sulfadoxine in combination with pyrimethamine is used as an antimalarial drug
- Nocardiosis: is an infectious disease caused by **Nocardia** that classically manifests as an opportunistic infection in immunocompromised patients

# Trimethoprim

Trimethoprim, is a potent inhibitor of **bacterial dihydrofolate reductase**, its bacteriostatic alone but in combination with Sulfa drugs it become bactericidal

Trimethoprim most often compounded with sulfamethoxazole producing the combination called cotrimoxazole and producing synergistic action



**Mechanism of action:** The active form of folate is the tetrahydrofolate derivative. Trimethoprim, selectively inhibits bacterial dihydrofolic acid reductase, which converts dihydrofolic acid to tetrahydrofolic acid which to leads to the synthesis of purines and ultimately to DNA.

Trimethoprim is about 50,000 times less efficient in inhibition of mammalian dihydrofolic acid reductase.

**Resistance:** to trimethoprim can result from

- Decreased cell permeability or active efflux pump
- Overproduction of dihydrofolate reductase or production of an altered dihydrofolate reductase that has a lower affinity for trimethoprim.

## Pharmacokinetics:

Trimethoprim is rapidly absorbed following oral administration. The drug is widely distributed into body tissues and fluids, including penetration into the CSF.

- Trimethoprim undergoes some O-demethylation, but 60% - 80% is renally excreted unchanged.
- Trimethoprim (a weak base) concentrates in prostatic fluid and in vaginal fluid, which are more acidic than plasma. Therefore, it has higher concentrations and more antibacterial activity in prostatic and vaginal fluids than many other antimicrobial drugs.

## Adverse effects:

Trimethoprim can produce folic acid deficiency, and Leads to megaloblastic anemia, leukopenia, and granulocytopenia, may need folic acid supplement

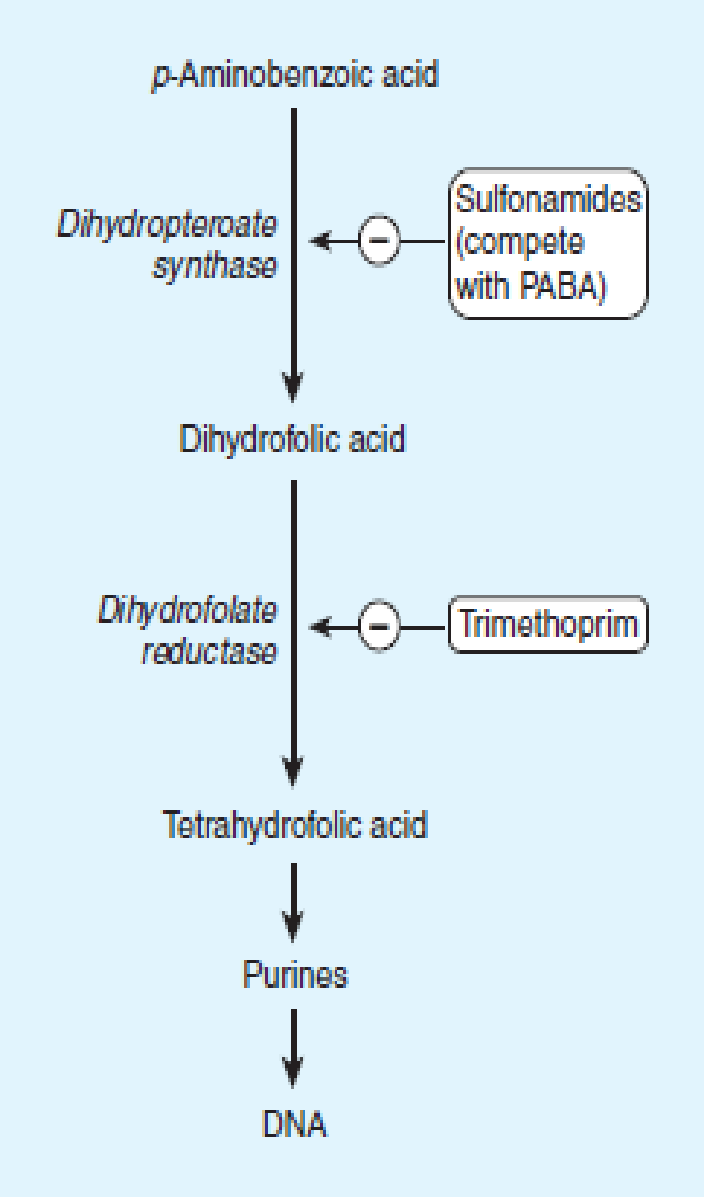
# Cotrimoxazole

The combination of **trimethoprim** with **sulfamethoxazole**, called **cotrimoxazole** shows greater antimicrobial activity, it is 20-100x more potent than either drugs used alone. It is bactericidal and has broad spectrum antibacterial activity.

## Mechanism of action:

The synergistic antimicrobial activity of cotrimoxazole results from its inhibition of two sequential steps in the synthesis of tetrahydrofolic acid.

Sulfamethoxazole inhibits the incorporation of PABA into dihydrofolic acid precursors, and trimethoprim prevents reduction of dihydrofolate to tetrahydrofolate



**Resistance:** to the trimethoprim–sulfamethoxazole combination is less frequently encountered than resistance to either of the drugs alone.

**Pharmacokinetics:** Cotrimoxazole is generally administered orally. IV administration may be used in patients with severe pneumonia.

Both agents distribute throughout the body, Cotrimoxazole readily crosses the blood–brain barrier. Both parent drugs and their metabolites are excreted in the urine.

**Adverse effects:** same as sulfa drugs alone in addition to the following:

Nausea and vomiting  are very common

Glossitis and stomatitis

Hyperkalemia

Megaloblastic anemia, leukopenia, thrombocytopenia reversed by the concurrent administration of **folinic acid.**

Hemolytic anemia in patients with G6PD deficiency

Increased INR in patients receiving both **sulfamethoxazole** and **warfarin**

## Clinical Indications:

- is effective for community-acquired MRSA, skin and soft tissue infections.
- Respiratory infections( effective against H. influenzae and Legionella pneumophila).
- Uncomplicated UTIs
- Pneumocystis jirovecii pneumonia (opportunistic infection in AIDS pts).
- Septicemia and meningitis caused by Listeria monocytogenes.
- Gastrointestinal infections (Shigellosis, management of carriers of S. typhi).
- Toxoplasmosis

## Other sulfonamide combinations:

**Fansidar** is a combination of sulfadoxine + pyrimethamine is used as alternative treatment of malaria in chloroquine –resistant Plasmodium falciparum

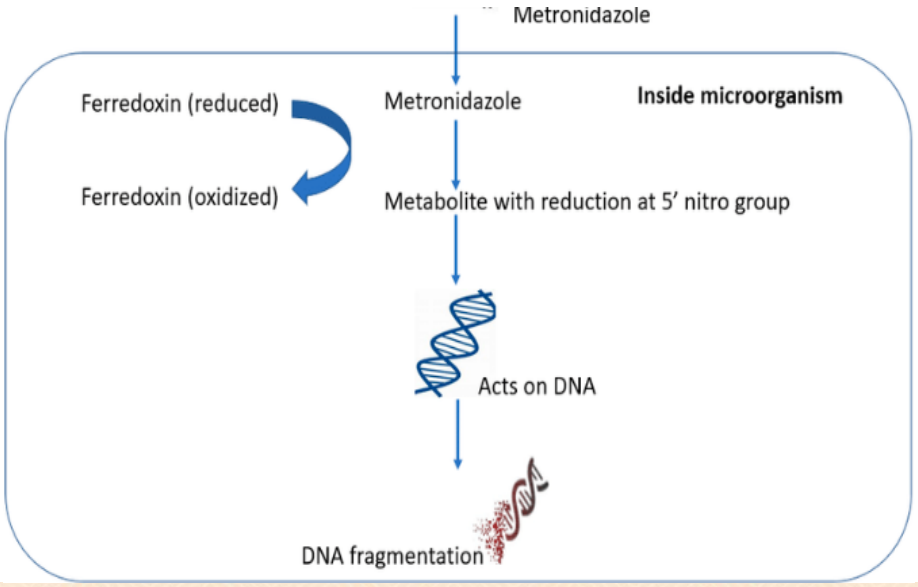
**Sulfasalazine** is a combination of sulfapyredine + 5-aminosalicylic acid

Used in treatment of ulcerative colitis.

**Silver sulfadiazine** is used topically for the treatment of burn.

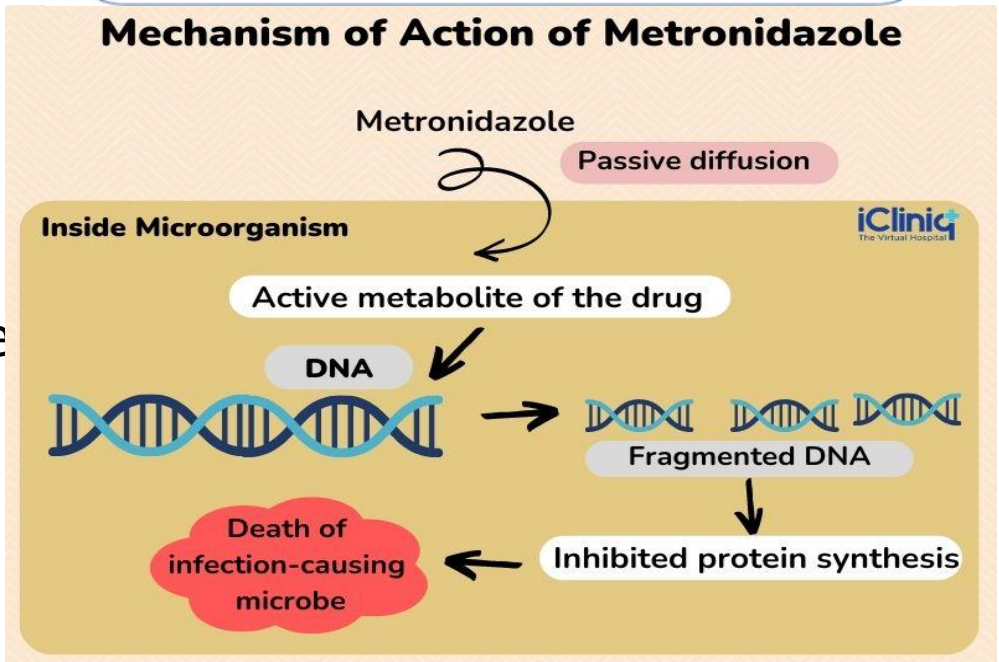
# Metronidazole

**Metronidazole** is a synthetic drug with high activity against anaerobic bacteria such as Bacteroids, Clostridia and some streptococci, . It is also active against several medically important protozoa and parasites.




## Mechanism of action:

Non-enzymatic reduction of nitro group by the organism leads to production of toxic free radical metabolites in the bacterial cell that damage DNA and other macromolecules



**Pharmacokinetics:**

Metronidazole is well absorbed after oral or rectal administration, but is often administered by IV route. The half-life is approximately 6 hrs. It is eliminated by a combination of hepatic metabolism and renal excretion.

**Adverse effects:** These include:

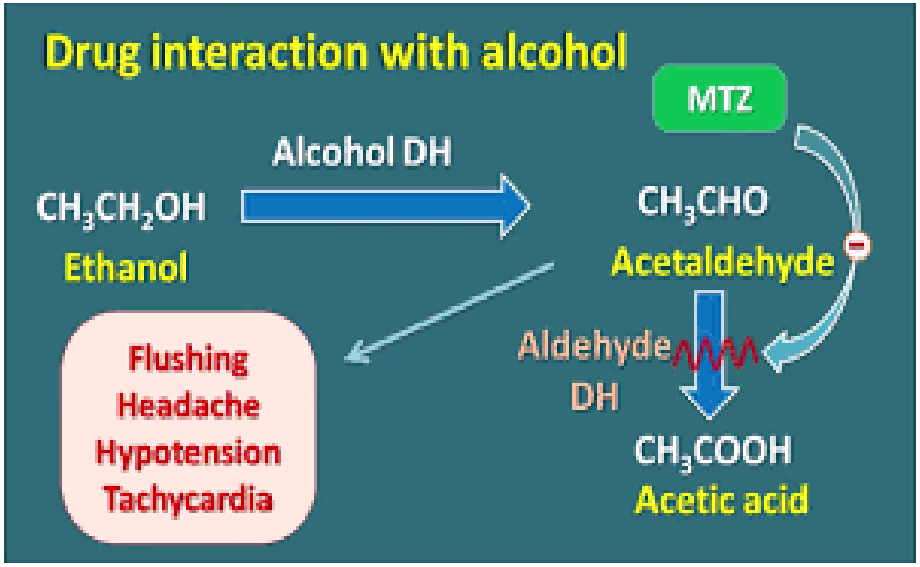
nausea and vomiting;

peripheral neuropathy;

convulsions, headaches;

hepatitis.

Metallic taste




**Drug interactions:**

Metronidazole interacts with alcohol because it inhibits aldehyde dehydrogenase and consequently causes a disulfiram like reaction. It is a weak inhibitor of cytochrome P450

## Clinical indications :

- It is used to treat trichomonal infections, amoebic dysentery, giardiasis, gas gangrene, pseudomembranous colitis
- Anaerobic or mixed intra-abdominal infections,
- Lung abscesses,
- Dental sepsis (periodontitis)
- Peptic ulcer (in combination with PPI and clarithromycin)
- It is used prophylactically before abdominal surgery.